

# Combinatorics – 6

## Combinatorial geometry

### Homework

14 сентября 2026 года

**Theoretical minimum.** A finite set of points is in general position if no three are collinear and in convex position if all are vertices of the convex hull. Among any five points in general position four are in convex position, and for every  $n$  there is  $N(n)$  such that  $N(n)$  points in general position contain  $n$  in convex position (Erdős–Szekeres, from Ramsey’s theorem of Combinatorics – 5); the bounds are  $2^{n-2} + 1 \leq N(n) \leq \binom{2n-4}{n-2} + 1$ . The unit-distance graph contains no  $K_{2,3}$ , since two circles meet twice, whence at most  $O(n^{3/2})$  unit distances and at least  $\Omega(\sqrt{n})$  distinct distances by double counting. The chromatic number of the plane lies between 5 and 7. Sylvester–Gallai is in Methods of proof – 3, Helly and Radon in Convex – 1, Minkowski’s lattice theorem and Pick’s formula in Convex – 10, and probabilistic colouring bounds in Probability – 4; all are used here without proof.

---

1. Let  $V$  be a finite set of points in the plane. Call  $V$  *balanced* if for any two distinct  $A, B \in V$  there is  $C \in V$  with  $AC = BC$ , and *center-free* if for any distinct  $A, B, C \in V$  no point  $P \in V$  satisfies  $PA = PB = PC$ . (a) Show that for every  $n \geq 3$  a balanced set of  $n$  points exists. (b) For which  $n \geq 3$  does a balanced center-free set of  $n$  points exist? (IMO shortlist 2015, C2)
2. Let  $n > 1$  and let  $2n$  points in the plane be given, no three collinear. The points are to be labelled  $A_1, \dots, A_{2n}$  in some order, and one considers the  $2n$  angles  $\angle A_1 A_2 A_3, \angle A_2 A_3 A_4, \dots, \angle A_{2n} A_1 A_2$ , each measured as the smaller positive value. Prove that there is a labelling for which these angles split into two groups of equal total. (IMO shortlist 2019, C6)

3. In the plane let  $D_1, \dots, D_n$  be pairwise disjoint disks with radii  $R_1 \geq R_2 \geq \dots \geq R_n$ ,  $n \geq 6$ , and let  $P_i \in D_i$ . Prove that for every point  $O$

$$OP_1 + OP_2 + \dots + OP_n \geq R_6 + R_7 + \dots + R_n.$$

(IMO shortlist 2020, G9)

4. Consider an  $N \times N$  grid. A right-down path is a sequence of cells each of which is one cell to the right of or one cell below the previous one; a right-up path is defined likewise. Prove that the cells cannot be partitioned into fewer than  $N$  such paths. (IMO shortlist 2023, C-problem)

5. A set of points in the plane is called *sane* if no three of them are collinear and the angle determined by any three distinct points is a rational number of degrees. (a) Does a countably infinite sane set exist? (b) Does an uncountable one? (ICMC)

6. A set of points in the plane is called *rigid* if every point is equidistant from the three (or more) points nearest to it. (a) Does a rigid set of 9 points exist? (b) Of 11 points? (ICMC)

7. Prove that there exist positive constants  $C$  and  $t > 1$  such that for  $n \geq 2$ , any  $2n$  points in general position consisting of  $n$  red and  $n$  blue points determine at least  $Cn^t$  triangles with two red vertices and one blue vertex whose interior contains no point of the set. (CIIM 2025)

8. Prove Monsky's theorem: a square cannot be cut into an odd number of triangles of equal area. Use the 2-adic valuation to three-colour the plane and Sperner's lemma to find a rainbow triangle, and explain which step fails for a rectangle of irrational aspect ratio.

9. Prove that the chromatic number of the plane is at least 4 by the Moser spindle and at most 7 by a hexagonal tiling. Then prove that a unit-distance graph on  $n$  vertices has  $O(n^{3/2})$  edges and deduce that it has a vertex of degree  $O(\sqrt{n})$ , hence is  $O(\sqrt{n})$ -colourable.

10. Prove Pick's theorem  $A = I + B/2 - 1$  for a lattice polygon, first for lattice triangles and then by triangulation, and deduce that a convex lattice polygon of area  $A$  has at most  $A + 2$  vertices. Then show that a convex lattice polygon with no lattice point other than its vertices has at most four vertices. (Minkowski's lattice theorem is Convex – 10 and Multiple integrals – 3, homework 8.)